

1. What is a Map in Java?

Ans: A Map is a collection in Java which stores data/information in the format of key-value pairs and each key is unique.

It is written in code as:

```
Map<Key, Value> map = new ....;
```

2. What are the commonly used implementations of Map in Java?

Ans:

The commonly used implementations of Map in Java are:

HashMap

TreeMap

LinkedHashMap

ConcurrentHashMap

3. What is the difference between HashMap and TreeMap?

Ans:

HashMap

- HashMap is an unordered collection of elements.
- It uses the hashing concept to store Key-Value pairs in an unordered manner.

TreeMap

- TreeMap is an ordered, sorted collection of elements
- It uses a sorted ordered to store Key-Value pairs based on the natural ordering of keys.

4. How do you check if a key exists in a Map in Java?

Ans:

There are two ways to check if a key exists in a Map in Java. One is by using the get() method which returns the value corresponding to the key if it exists in the Map or returns a null if it does not exist.

The other way is to use the containsKey() method which returns a boolean value True if the key exists in the Map and False if it does not exist.

5. What are Generics in Java?

Ans:

Generics is a way of providing type safety to the collections in Java by allowing the use of generic types. Since Java is a strongly typed language, it is recommended to use generics <Type> along with the collection declaration.

It also reduces the code redundancy as it allows classes, methods and interfaces to be written generically, without specifying the type of data being used.

6. What are the benefits of using Generics in Java?

Ans:

The benefits of using Generics in Java are:

- Type safety

- Code reusability
- Improved readability
- Reduced code redundancy
- Improved performance

7. What is a Generic Class in Java?

Ans:

A Generic Class in Java is a class that can work with different types of data. It is defined using a type parameter enclosed in angle brackets, which represents the type of data being used.

8. What is a Type Parameter in Java Generics?

Ans:

A Type Parameter in Java Generics is a placeholder for the type of data that is used by a generic class or method. It is defined using a single uppercase letter enclosed in angle brackets, such as <T> or <E>.

9. What is a Generic Method in Java?

Ans:

A Generic Method in Java is a method that can work with different types of data. It is defined using a type parameter enclosed in angle brackets, which represents the type of data being used.

10. What is the difference between ArrayList and ArrayList<T>?

Ans:

ArrayList is a non-generic class, while ArrayList<T> is a generic class. ArrayList<T> provides type safety, as it can only store elements of the specified type, whereas ArrayList can store any type of element.